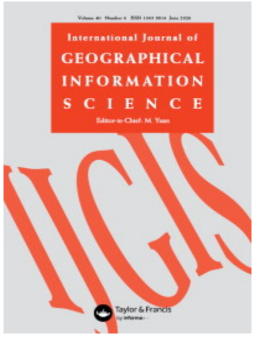

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RESEARCH ARTICLE

USGT: an unsupervised spatial graph transformer for detecting spatial communities in human mobility data

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ABSTRACT

Spatially embedded graphs derived from human mobility data are usually sparse and noisy, making it difficult for existing statistical methods to detect underlying spatial communities. Graph embedding techniques have been used to obtain low-dimensional representations for community detection; however, they often fail to preserve modular structure and rarely incorporate spatial constraints inherent in spatially embedded graphs. To overcome these limitations, we proposed an Unsupervised Spatial Graph Transformer (USGT) for spatial community detection. To incorporate spatial constraints into the model, we designed a spatial encoding scheme that explicitly encodes spatial proximity between spatial units. To preserve the modular structure in graphs, we developed a structure-aware self-attention mechanism that captures both global and local information. A data masking strategy was further employed to facilitate unsupervised learning. Experiments on synthetic datasets, evaluated using Adjusted Rand Index and Normalized Mutual Information, demonstrated that USGT outperformed six state-of-the-art methods. A case study on taxi GPS trajectory data in Beijing further showed that the spatial communities identified by USGT can enhance the understanding of mobility patterns and offer insights into the polycentric urban structure.

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KEYWORDS
Spatial community detection; graph embedding; transformer; self-attention mechanism; human mobility data

1. Introduction

The advancement of location-aware technologies has led to a significant increase in the availability of massive human movement data, such as mobile phone data (Yuan and Raubal 2016, Zhao *et al.* 2023), smart card data (Zhong *et al.* 2014), taxi GPS trajectory data (Liu *et al.* 2022, Wang *et al.* 2025), and bike-sharing data (Li *et al.* 2020). These data offer critical insights into how people interact with the urban environment, which is helpful for uncovering the underlying structure of urban space (Jia *et al.* 2022). If a city is considered to be composed of geographical units, movement data

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can be represented as a spatially embedded weighted graph, where nodes correspond to spatial units (e.g. traffic analysis zones, TAZs) and weighted edges represent interactions between units, such as origin–destination (OD) pairs (Batty 2013). In such graphs, spatial community structures are defined as spatially contiguous regions where interactions within the region are denser than those between the region and other regions (Guo *et al.* 2018). Identifying such spatial community structures can provide valuable insights for urban planning (Liu *et al.* 2015) and traffic management (Lu *et al.* 2018).

However, the spatially embedded networks derived from movement data are often sparse and noisy (Xu *et al.* 2007, Barabási and Pósfai 2016), which makes the accurate identification of spatial communities challenging. In recent years, a number of statistical methods have been developed for spatial community detection (Chen *et al.* 2015, Wang *et al.* 2021, Liu *et al.* 2025); however, they predominantly rely on the statistical characteristics of interactions. Therefore, detecting spatial communities from sparse movement data remains a challenge. Recently, graph embedding techniques, such as DeepWalk (Perozzi *et al.* 2014) and node2vec (Grover and Leskovec 2016), have attracted increasing attention. These methods transform complex networks into low-dimensional embeddings that preserve essential structural properties, thereby providing new opportunities for detecting communities in sparse networks (Kojaku *et al.* 2024). However, most existing techniques are not specifically designed for community detection; thus, applying them directly does not guarantee better performance (Tandon *et al.* 2021). Moreover, the spatial characteristics of spatially embedded graphs are rarely considered during embedding, which may cause the embeddings to overlook spatial proximity and yield fragmented community structures.

To address these limitations, we developed an Unsupervised Spatial Graph Transformer (USGT) framework for detecting spatial communities from sparse and noisy human mobility data. The primary contributions of this study are as follows: (1) We modified the self-attention mechanism to integrate both global and local node information, which is crucial for preserving modular structure. (2) We designed a novel spatial encoding scheme and integrated it into the model to capture spatial relationships between nodes, which are derived from the spatial proximity between spatial units. (3) We generated multiple subgraphs using a data masking strategy to facilitate model training, and applied an encoder-decoder structure to enable unsupervised learning.

Simulation experiments showed that the proposed method identified spatial communities more accurately and completely than six representative methods. Its application to taxi trajectory data in Beijing further revealed the city's polycentric structure.

The remainder of the paper is organized as follows. Section 2 reviews related work, Section 3 outlines the proposed methodology, Section 4 reports the simulation experiments, Section 5 presents the case study, Section 6 provides a discussion, and Section 7 concludes the study.

2. Literature review

Existing spatial community detection approaches fall into two categories: (i) statistical approaches and (ii) graph embedding-based approaches. Statistical approaches partition the graph by leveraging the statistical characteristics of interactions. In contrast, graph embedding approaches first transform nodes into a low-dimensional vector space and then employ clustering methods to identify community structures.

2.1. Statistical methods for spatial community detection

Early studies directly applied community detection algorithms originally developed for complex networks to identify spatial communities (Zhong *et al.* 2014, Liu *et al.* 2015). Among these, objective optimization-based methods are the most widely adopted for spatial community detection. Specifically, modularity is one of the most widely used objective functions in spatial community detection. It measures the difference between the observed number of connections within communities and the expected number of connections under a null model (Newman and Girvan 2004). However, the direct application of such methods may lead to the identification of spurious communities (e.g. spatially fragmented communities) (Guo *et al.* 2018).

To address these limitations, dedicated approaches for spatial community detection have emerged, generally falling into two categories according to their underlying strategies:

1. Methods that modify the objective function by incorporating spatial effects. For instance, Expert *et al.* (2011) modified the null model of modularity by incorporating a gravity model to estimate the expected network connections. Similarly, Gao *et al.* (2013) proposed a new modularity definition by comparing observed connectivity with expected connectivity derived from a gravity model. Chen *et al.* (2015) incorporated distance into the modularity, assigning connection weights according to the inverse of spatial distance to the power of n. While these methods are effective for specific purposes, they introduce extra parameters (e.g. configurations of the distance decay model) and assumptions (e.g. distance decay).
2. Methods that incorporate spatial constraints to the optimization process. For instance, Guo *et al.* (2018) proposed a spatially constrained Tabu optimization framework to detect spatial community structures in movements. Wang *et al.* (2021) refined the Louvain (Blondel *et al.* 2008) and Leiden (Traag *et al.* 2019) methods by incorporating a spatial contiguity constraint to identify spatially contiguous communities. Cai *et al.* (2025) integrated spatial constraints into the Infomap algorithm (Rosvall and Bergstrom 2008) by applying spatial Tabu optimization for community structure (Guo *et al.* 2018) to optimize the map equation. These methods have demonstrated greater robustness and effectiveness in uncovering spatial community structures.

2.2. Graph embedding methods for spatial community detection

Graph embedding is an important technique for learning low-dimensional node embeddings, with the aim of preserving the underlying structure of the graph. Current graph embedding methods can be primarily categorized as follows.

1. Matrix factorization-based methods: These methods employ the eigenvectors of graph matrices to project nodes into a low-dimensional vector space. For instance, Belkin and Niyogi (2003) developed the Laplacian eigenmap algorithm, which embeds more

densely connected nodes closer together by minimizing an objective function. To retain high-order proximity between nodes in the embedding space, Ou *et al.* (2016) further developed the higher-order proximity preserving embedding (HOPE) method based on non-negative matrix factorization. However, these methods suffer from both computational and statistical limitations, making it challenging to scale them for large networks (Tang *et al.* 2015, Grover and Leskovec 2016).

2. Random walk-based methods: These methods frame node embedding learning as analogous to word embedding learning in natural language processing, with truncated random walks considered as sentences. DeepWalk (Perozzi *et al.* 2014), a widely recognized technique in this category, exploits local information captured by truncated random walks to learn node embeddings through the skip-gram model (Mikolov *et al.* 2013). Building on DeepWalk, node2vec (Grover and Leskovec 2016) introduces a biased random walk procedure to add flexibility in neighborhood exploration, which enables the learning of richer representations. Despite their success, these approaches rely on shallow models. Given that the underlying graph structures are often highly non-linear (Luo *et al.* 2011), such models may fail to capture these complexities, leading to sub-optimal representations (Wang *et al.* 2016).
3. Deep neural network-based methods: To effectively model the non-linear dependencies of graph structures, these methods employ deep neural networks. For instance, graph convolutional networks (GCN, Kipf and Welling 2017) and graph attention networks (GAT, Veličković *et al.* 2018) can aggregate information from neighboring nodes to encode local structural information into node embeddings. Specifically, autoencoders (AEs, Hinton and Salakhutdinov 2006) are commonly used for community detection, as label information is typically unavailable in this task. Wang *et al.* (2016) developed a semi-supervised deep model based on an autoencoder architecture. This model simultaneously optimizes first- and second-order similarities to learn node representations. To further incorporate prior knowledge of node memberships, Yang *et al.* (2016) proposed a nonlinear reconstruction algorithm based on deep neural networks, which integrates pairwise constraints between adjacent nodes into the autoencoder loss function.

While the above methods effectively capture structural dependencies in graphs, they rarely account for spatial factors. More recent studies have incorporated spatial relationships by constructing spatial adjacency matrices based on geographical contiguity, which are then used as the underlying graph structure for graph convolutional networks (Liang *et al.* 2022, Liang *et al.* 2025). In these approaches, spatial proximity is treated as a structural prior that constrains information propagation, enabling graph neural networks to capture local spatial dependencies.

2.3. Detection of spatial communities: a critical analysis and new strategy

Despite notable progress, existing approaches to spatial community detection still face significant challenges. Statistical methods primarily rely on the statistical characteristics of interactions, making it difficult to handle sparse and noisy movement data. Moreover, these methods may detect spurious communities even in random graphs

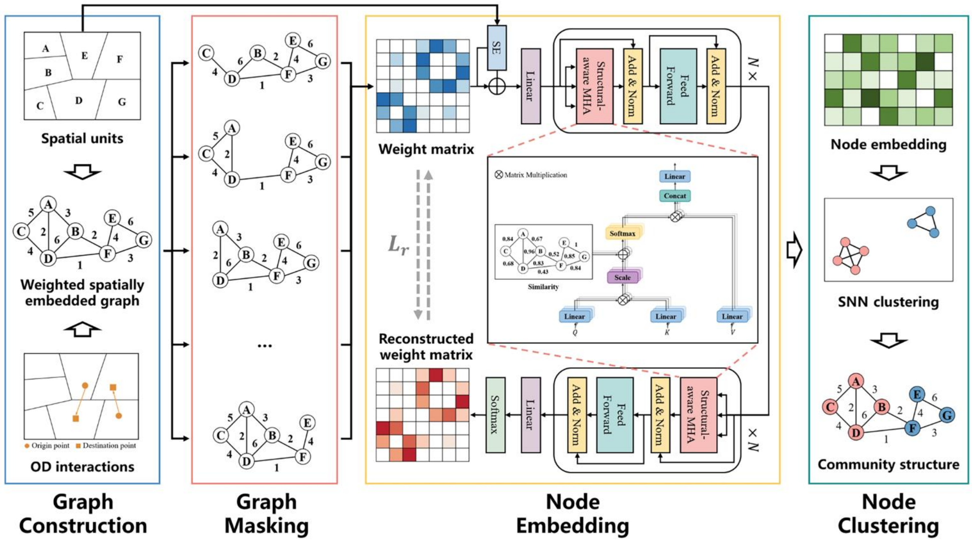


Figure 1. Framework of the proposed method.

without any true underlying community structure (Zhang and Moore 2014). Graph embedding techniques project sparse graphs into low-dimensional vector spaces, offering the potential to uncover latent community structures (Su *et al.* 2024). However, most existing graph embedding techniques are not specifically designed for community detection. Preserving community structures during the embedding process remains a significant challenge (Tandon *et al.* 2021).

Recently, graph Transformers have emerged as powerful models for graph-structured data, enabling the modeling of long-range dependencies (Kreuzer *et al.* 2021, Rampášek *et al.* 2022). This capability is particularly important for sparse graphs. In such graphs, nodes within the same community may not be directly connected, requiring the model to infer associations between structurally related yet topologically disconnected nodes. However, graph Transformers may suffer from the problem of over-globalization, where the attention mechanism tends to emphasize topologically distant nodes while overlooking neighboring nodes that may contain more valuable information (Xing *et al.* 2024). In addition, applying graph Transformers to spatially embedded graphs poses a fundamental challenge: how to preserve both topological and spatial relationships between nodes. Furthermore, given the large number of parameters in Transformer models, training such models in an unsupervised manner remains challenging. To overcome these limitations, we proposed a novel framework based on deep graph embedding for detecting spatial communities in movement data (Figure 1).

To capture both interaction intensities and spatial relationships between nodes, we constructed a spatially constrained interaction graph based on OD pairs and spatial relationships among spatial units. Specifically, interactions between nodes were encoded using a weighted adjacency matrix, while spatial relationships among spatial units were modeled using a spatial encoding scheme. The final model input was obtained by combining the weighted adjacency matrix with the spatial encoding.

To capture complex dependencies in spatially embedded graphs, we developed an encoder-decoder architecture based on the vanilla Transformer (Vaswani *et al.* 2017), enhanced with a structure-aware multi-head self-attention (SMHA) mechanism. Specifically, to alleviate the over-globalizing issue in graph Transformers, SMHA incorporated the strength of connections between node pairs into the attention computation, thereby preserving both global and local structural information. The encoder/decoder layers consist of an SMHA component and a position-wise feed-forward network (FFN), each followed by residual connections and layer normalization.

Furthermore, to enable unsupervised training, we employed a data masking strategy and minimized a composite loss function. Finally, as the learned node embeddings are high-dimensional and non-uniform, we applied the shared nearest neighbor (SNN) clustering method (Ertöz *et al.* 2003) to identify spatial community structures.

3. Methodology

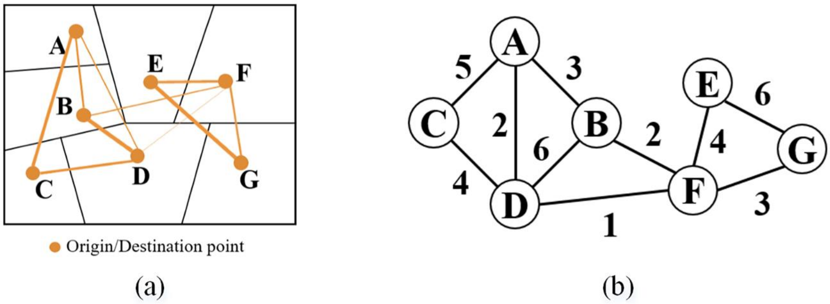
3.1. Construction of spatially constrained interaction graph

To incorporate both interactions and spatial relationships between nodes, we generated a spatially constrained interaction graph based on OD pairs and the spatial relationships among spatial units. A spatially embedded graph is defined as $G = \{\boldsymbol{V}, \boldsymbol{E}, \boldsymbol{W}\}$, where $\boldsymbol{V}$ denotes the set of spatial units, $\boldsymbol{E}$ consists of edges formed by OD pairs, and $\boldsymbol{W}$ represents the set of edge weights, with each weight w_{ij} representing the number of OD pairs between nodes i and j. Self-loops are not considered in this study; thus, $w_{ij} = 0$ when $i = j$. Figure 2 illustrates an example of constructing a spatially embedded graph. Specifically, Figure 2(c) shows the corresponding weighted adjacency matrix $\boldsymbol{M}^w$, where each entry $\boldsymbol{M}^w_{ij}$ denotes the edge weight between nodes i and j. Each row of $\boldsymbol{M}^w$ serves as the initial representation of a node and is represented as an n-dimensional vector.

Although the weighted adjacency matrix encodes the strength of interactions, it does not capture spatial relationships among nodes, which are critical for identifying spatial communities. To address this limitation, we designed a spatial encoding scheme to integrate spatial information into the model. Instead of imposing spatial proximity as a hard constraint during community detection, we employed a soft strategy that does not restrict communities to spatially contiguous regions. We assumed that geographically proximate nodes exhibit stronger spatial dependencies. To encode such dependencies, we constructed a matrix $\boldsymbol{M}^{spa}$, in which each row corresponds to the spatial encoding (SE) of a node. Let $r = 1, 2, \ldots$ denote the rank of neighbors of a node based on their spatial proximity, with smaller values of r corresponding to closer neighbors. If node i is the r-th neighbor of node j, then the entry $\boldsymbol{M}^{spa}_{ij}$ is assigned a value of $1/r$. For instance, as illustrated in Figure 3, node C is a second-order neighbor of node A; thus, $\boldsymbol{M}^{spa}_{AC} = 1/2$. Finally, we combined the spatial encoding matrix with the weighted adjacency matrix to form the model input:

$$\boldsymbol{X} = \boldsymbol{M}^w + \alpha \boldsymbol{M}^{spa} \tag{1}$$

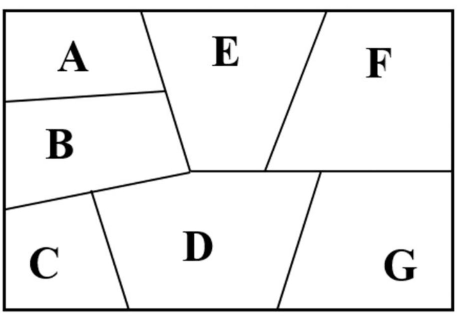
where α represents a learnable scalar parameter. This formulation enables the proposed model to leverage spatial relationships while preserving the flexibility to group spatially non-contiguous yet tightly connected nodes into the same community.



	A	B	C	D	E	F	G
A	0	3	5	2	0	0	0
B	3	0	0	6	0	2	0
C	5	0	0	4	0	0	0
D	2	6	4	0	0	1	0
E	0	0	0	0	0	4	6
F	0	2	0	1	4	0	3
G	0	0	0	0	6	3	0

(c)

Figure 2. Construction of a spatially embedded graph: (a) spatial units and OD interactions; (b) topological structure of the spatially embedded graph; (c) corresponding weighted adjacency matrix.



	A	B	C	D	E	F	G
A	0	1	0.5	0.5	1	0.5	0.33
B	1	0	1	1	1	0.5	0.5
C	0.5	1	0	1	0.5	0.5	0.5
D	0.5	1	1	0	1	1	1
E	1	1	0.5	1	0	1	0.5
F	0.5	0.5	0.5	1	1	0	1
G	0.33	0.5	0.5	1	0.5	1	0

(a) (b)

Figure 3. Spatial encoding of spatial units: (a) distribution of spatial units; (b) corresponding spatial encoding.

3.2. Structure-aware multi-head self-attention for spatial community detection

Although Transformers effectively capture long-range dependencies through global self-attention, they may overemphasize topologically distant nodes (Xing *et al.* 2024). This over-globalization can obscure localized community structures. In general, preserving community structure during the embedding process remains a challenge for

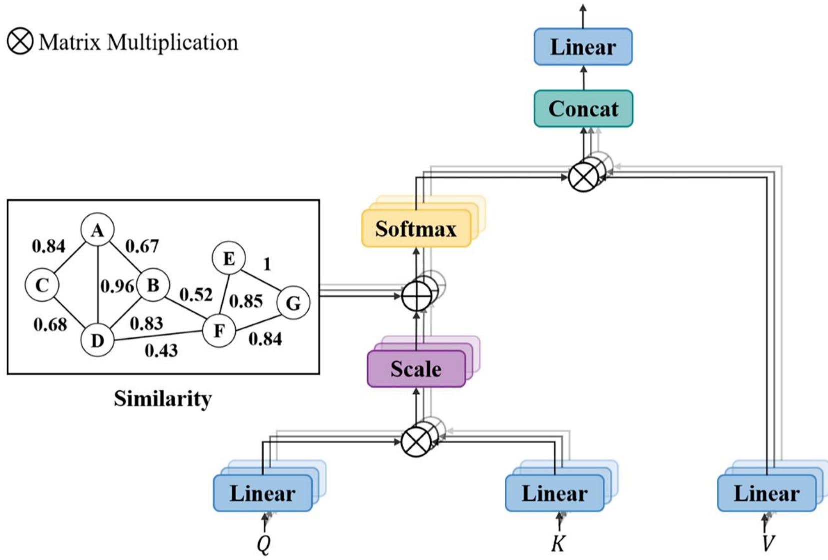


Figure 4. Structure-aware multi-head self-attention module.

existing embedding-based methods. Approaches that emphasize local proximity (e.g. neighboring nodes) often fail to capture meso-scale community patterns beyond immediate neighborhoods, resulting in fragmented community representations. Conversely, methods that rely solely on global interactions may dilute intra-community connectivity by overemphasizing topologically distant nodes.

To address this challenge, we developed a structure-aware multi-head self-attention mechanism that incorporates local structural similarity between node pairs into the attention computation (Figure 4). Since nodes sharing more neighbors are more likely to belong to the same community, we quantified the local structural similarity between directly connected node pairs to guide the attention mechanism. This design encourages the model to focus on structurally similar neighbors and enhances its ability to preserve community structure in the learned embeddings. For each directly connected node pair (i, j), the structural similarity $\mathbf{S}_{ij}$ is computed based on shared neighbors as follows:

$$\mathbf{S}_{ij} = \frac{\sum_{x \in N(i) \cap N(j)} [w_{ix} + w_{jx}]}{\sum_{x \in N(i) \cup N(j)} [w_{ix} + w_{jx}]} \tag{2}$$

where $N(i)$ denotes the set consisting of node i and all nodes directly connected to it, and w_{xy} represents the edge weight between nodes x and y. The similarity score is computed only for topologically connected node pairs. For nodes i and j that are not directly connected, $\mathbf{S}_{ij}$ is set to 0.

Given the input node features $\mathbf{H} \in \mathbb{R}^{n \times d}$, the queries, keys, and values for each attention head h are generated via linear projections:

$$\mathbf{q}_i^{(h)} = \mathbf{H}_i \mathbf{W}_Q^{(h)}, \mathbf{k}_j^{(h)} = \mathbf{H}_j \mathbf{W}_K^{(h)}, \mathbf{v}_j^{(h)} = \mathbf{H}_j \mathbf{W}_V^{(h)} \tag{3}$$

where $\mathbf{W}_Q^{(h)}$, $\mathbf{W}_K^{(h)}$ and $\mathbf{W}_V^{(h)} \in \mathbb{R}^{d \times d_h}$ denote the parameter matrices for attention head h. The attention score for each node pair (i, j) is computed as:

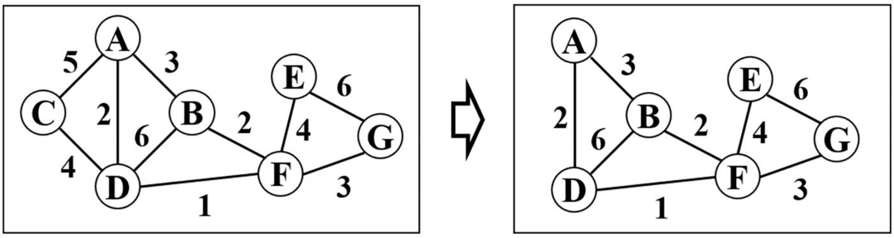


Figure 5. An example of data masking.

$$\alpha_{ij}^{(h)} = \text{softmax}\left(\frac{\mathbf{q}_i^{(h)} \cdot \mathbf{k}_j^{(h)}}{\sqrt{d_h}} + \beta \mathbf{S}_{ij}\right) \tag{4}$$

where β is a learnable scalar parameter. This formulation encourages each attention head to emphasize directly connected and structurally similar nodes, while retaining the Transformer's ability to capture global dependencies. The output of attention head h for node i is then given by:

$$\boldsymbol{a}_i^{(h)} = \sum_{j \in V} \alpha_{ij}^{(h)} \cdot \mathbf{v}_j^{(h)} \tag{5}$$

Finally, the outputs from multiple attention heads are concatenated, followed by a linear projection:

$$\text{SMHA}(\mathbf{H})_i = \text{Concat}\left(\boldsymbol{a}_i^{(1)}, \ldots, \boldsymbol{a}_i^{(n_h)}\right)\mathbf{W}_O \tag{6}$$

where n_h represents the number of attention heads, and $\mathbf{W}_O \in \mathbb{R}^{n_h d_h \times d}$ projects the concatenated output back to the original feature dimension.

3.3. Unsupervised model learning via a data masking strategy

A large number of samples is essential for effective model training. However, in community detection tasks, only a single target graph is usually available. To address this limitation, we employed a data masking strategy, which assumes that a sufficiently large number of subgraphs can approximate the distribution of the entire graph. Specifically, a certain percentage of nodes and their associated edges were randomly removed from the target graph, as shown in Figure 5. We define the data masking rate as the proportion of discarded nodes and analyze its impact on model performance in Section 4.4.3. Training the proposed model on subgraphs improves computational efficiency, as processing the entire graph can be computationally expensive. Moreover, the random sampling of diverse subgraphs enables the model to learn more robust node representations and reduce the risk of overfitting.

To effectively train the model in an unsupervised manner, we defined a composite loss function that guides both the reconstruction of the input graph and the preservation of underlying node relationships. Specifically, the reconstruction loss L_r quantifies how accurately the model reconstructs the weighted adjacency matrix:

$$L_r = MSE\left(\mathbf{W}, \mathbf{W}'\right) \tag{7}$$

where $\mathbf{W}$ denotes the weighted adjacency matrix of the input graph, and $\mathbf{W}'$ denotes the reconstructed matrix generated by the model.

While L_r supports unsupervised learning by enforcing graph reconstruction, it does not explicitly preserve relationships between nodes in the embedding space. To address this, we introduced a structure-preserving loss L_s, which measures the difference between node-pair similarities before and after embedding:

$$L_s = MSE(\mathbf{S}, \mathbf{S}') \tag{8}$$

where $\mathbf{S}$ is the structure-aware similarity matrix defined in Equation (2), and $\mathbf{S}'$ is computed from the learned node embeddings, where each entry $\boldsymbol{S}'_{ij}$ is computed as the inner product between the embedding vectors of nodes i and j.

The overall loss function combines Equations (7) and (8):

$$L = L_r + \lambda L_s \tag{9}$$

where λ is a balance coefficient that determines the trade-off between the two losses. The model parameters were optimized by minimizing the overall loss in Equation (9) using the Adam optimizer.

3.4. Node embeddings clustering

The obtained node embeddings are high-dimensional and non-uniform. To address these challenges, we employed the shared nearest neighbor (SNN) clustering algorithm (Ertöz *et al.* 2003), which is well-suited for high-dimensional and non-uniformly distributed data. SNN clustering is a density-based method that redefines similarity according to the number of shared nearest neighbors. The procedure can be summarized as follows:

1. Construction of the SNN graph: We first computed pairwise similarities between node embeddings. For each node, only the k most similar nodes were retained to construct a k-nearest neighbor graph. The SNN similarity between any two nodes is defined as the number of shared neighbors within their respective k-nearest neighbors.
2. Identification of core nodes: The density of a node is defined as the number of its k-nearest neighbors whose SNN similarity with the node exceeds a predefined threshold ε. A node is considered a core node if its SNN density is greater than $k/2$.
3. Generation of clusters: Clusters are formed by connecting core nodes whose SNN similarity exceeds ε. Each connected component of core nodes constitutes a cluster.
4. Assignment of non-core nodes: Non-core nodes whose SNN similarity with at least one core node exceeds ε are assigned to the cluster of their most similar core node. Nodes that are neither core nodes nor non-core nodes are treated as noise or outliers and are excluded from the final clustering result.

The SNN clustering method involves two key parameters: the number of nearest neighbors k and the threshold of the shared nearest neighbors ε. In this work, the similarity between node embeddings was measured using cosine similarity.

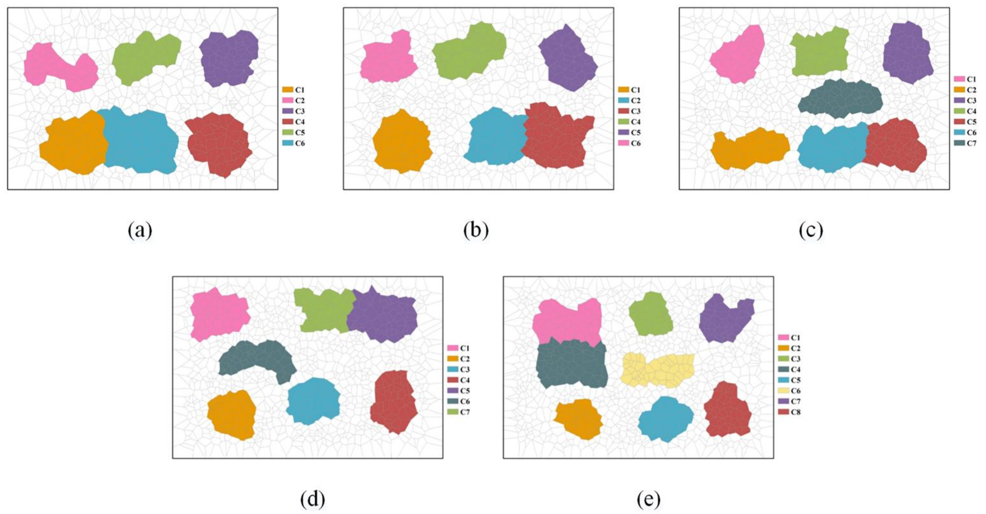


Figure 6. Five simulated datasets: (a) $N = 700$; (b) $N = 800$; (c) $N = 900$; (d) $N = 1000$; (e) $N = 1100$.

4. Simulation experiments

To evaluate the performance of the proposed method, we generated a series of simulated datasets and compared it with six representative methods: Louvain (Blondel *et al.* 2008), Leiden (Traag *et al.* 2019), ScLeiden (Wang *et al.* 2021), HOPE (Ou *et al.* 2016), node2vec (Grover and Leskovec 2016), and GAE (Kipf and Welling 2016). Specifically, Louvain and Leiden are classic modularity-based community detection algorithms. ScLeiden extends the Leiden algorithm by incorporating spatial constraints to identify spatially contiguous communities. These three methods are statistical approaches for spatial community detection. In contrast, HOPE, node2vec, and GAE are embedding-based approaches. HOPE is a representative matrix factorization-based method, whereas node2vec is a widely used random walk-based method.

In addition, we included a graph autoencoder (GAE)-based approach for comparison, following the framework proposed by Kipf and Welling (2016). Unlike the original GAE, which employs a GCN encoder and an inner-product decoder, our implementation uses a GAT encoder and an MLP-based decoder to reconstruct the weighted adjacency matrix. Specifically, for each pair of nodes, their latent representations are concatenated and fed into a multilayer perceptron (MLP) to predict the corresponding edge weight, enabling the reconstruction of weighted relationships between nodes. The GAE model was pre-trained using the strategy described in Section 3.3.

4.1. Generation of simulated datasets

A series of synthetic datasets with predefined spatial communities were generated following procedures similar to those in Karrer and Newman (2011) and Liu *et al.* (2022). As shown in Figure 6, five groups of simulated datasets were generated, with the number of polygons $N \in \{700, 800, 900, 1000, 1100\}$. Each group consisted of 10 datasets. In this study, a simulated dataset containing m predefined spatial communities was constructed in two steps as follows:

1. Generation of m predefined spatial communities: For each spatial community C, the community size n_c was randomly drawn from the range [30, 70]. A polygon was randomly selected and assigned to C, after which spatially adjacent polygons were iteratively added until the size of C reached n_c. This process was repeated m times to generate all predefined spatial communities.
2. Generation of interactions between nodes: First, $pN(N-1)$ OD pairs were randomly generated as background interactions in the study area. Then, for each spatial community with n_i nodes, $qn_i(n_i-1)$ OD pairs were randomly generated within the community. This ensured denser interactions within communities, with q controlling the detectability of the predefined community structure. As q increases, the community structure becomes more detectable. More importantly, it allows the number of OD pairs to vary with community size. In this study, we set $p = 0.03$ and $q = 0.12$.

4.2. Evaluation metrics and parameter settings

The performance of the six methods was evaluated using two widely adopted metrics: Adjusted Rand Index (ARI) (Hubert and Arabie 1985) and Normalized Mutual Information (NMI) (Tan *et al.* 2006). ARI assesses the consistency between two partitions by comparing node pairs and their community assignments. ARI ranges from −1 to 1, where higher values indicate more accurate community partitions. NMI quantifies the similarity between two partitions based on their mutual information. NMI values lie between 0 and 1, where higher values indicate better agreement with the ground-truth community structure.

For Louvain, Leiden, and ScLeiden, the resolution parameter γ was set to its default value (i.e. 1). In addition, for ScLeiden, the minimal region size was set to 15. For all embedding-based methods, the embedding dimension d was set to 128. HOPE does not require additional parameters. The remaining parameters were configured as follows.

1. node2vec: The window size was set to 10, the length of random walk to 40, the number of walks per node to 80, and the biased walk weights p and q were both set to 1.
2. GAE: The model was implemented using a GAT encoder with four attention heads. During training, 2000 subgraphs were randomly generated with a masking rate of 0.5. The model was trained for 40 epochs using the Adam optimizer with a learning rate of 0.0001.
3. USGT: The model was configured with 8 attention heads, 4 encoder/decoder layers, and a hidden layer dimension of 2048 in the FFN module. Additionally, 2000 subgraphs were generated with a masking rate of 0.5 and a balance coefficient λ of 1. The training was performed for 40 epochs with a learning rate of 0.0001.

Since node2vec was originally designed for unweighted graphs, whereas this study focuses on weighted graphs, we extended node2vec to a weighted version: when generating random walk sequences, the next node was selected with probability proportional to the weight of the connecting edge. Furthermore, for all embedding-based methods, to reduce the influence of clustering algorithm parameters, we evaluated

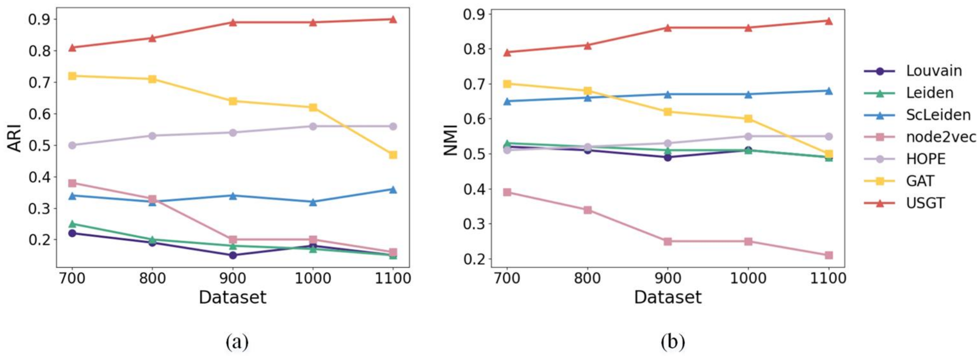


Figure 7. Performance of different methods on the simulated datasets: (a) ARI; (b) NMI.

multiple parameter combinations and reported the best-performing results for comparison. Specifically, the number of nearest neighbors k was varied from 5 to 40 in increments of 5, and the threshold of the shared nearest neighbors ε ranged from $k/4$ to $2k/3$. All parameter settings were selected from the same predefined ranges, and the best result among all parameter combinations was reported for each method.

4.3. Results and analysis

Figure 7 illustrates the performance of the seven methods on the simulated datasets. For each dataset, 10 networks were generated. We evaluated the detected community structures for each network individually and computed the corresponding ARI and NMI values. The final results were reported as the average ARI and NMI values over the 10 networks for each dataset. Overall, the proposed method consistently outperformed all comparative methods, achieving the highest ARI and NMI values across all simulated datasets. Among the baseline methods, Louvain and Leiden, both modularity-based algorithms, exhibited comparable performance, with Leiden slightly outperforming Louvain, likely due to differences in the optimization process, despite their similar objectives. ScLeiden, which extends Leiden by incorporating spatial constraints, achieved higher NMI values than the other modularity-based methods, indicating an improved ability to capture spatial community structures.node2vec performed relatively poorly, particularly in terms of NMI, indicating a limited capability to detect communities in noisy graphs. HOPE maintained relatively stable performance across different dataset sizes. Although HOPE outperformed Louvain, Leiden, and node2vec, it still performed worse than the proposed method. GAE achieved moderate performance on smaller datasets; however, its performance declined as dataset size increased, suggesting reduced effectiveness on larger graphs. Collectively, these results demonstrated that USGT was more robust and effective for community detection on simulated datasets than the six comparative methods.

Figure 8 displays the identified spatial communities using different methods for the dataset shown in Figure 6(e). The proposed method identified the predefined communities more accurately and more completely. In contrast, both Louvain and Leiden were strongly affected by noise, resulting in the incorrect assignment of noise nodes to communities and consequently lower ARI and NMI values. ScLeiden generated

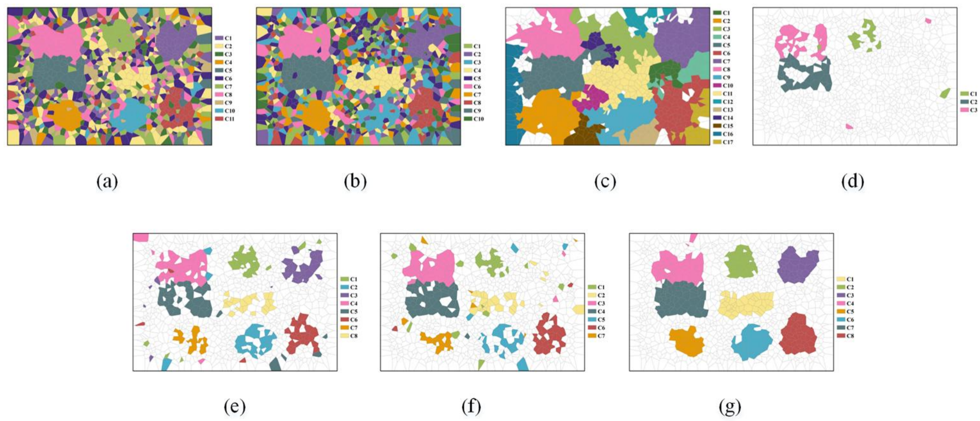


Figure 8. Spatial communities identified by different methods for a simulated dataset with $N = 1100$: (a) Louvain; (b) Leiden; (c) ScLeiden; (d) node2vec; (e) HOPE; (f) GAE; (g) the proposed method.

Table 1. Average running time of different methods on the simulated datasets (unit: seconds).

	700	800	900	1000	1100
Louvain	0.57	0.75	1.04	1.27	1.36
Leiden	0.08	0.12	0.13	0.19	0.21
ScLeiden	3.95	5.55	7.44	9.70	11.26
node2vec	323.59	412.72	519.41	639.05	766.74
HOPE	145.63	190.35	242.68	299.80	363.19
GAE	826.78	1021.48	1348.67	1493.73	1906.57
USGT	1315.51	1535.13	1785.39	2050.80	2359.89

spatially contiguous communities; however, it was also sensitive to noise, with some noise nodes incorrectly assigned and even forming spurious communities. node2vec identified only three of the predefined communities, and the detected community structures were incomplete, with some noise nodes mistakenly incorporated into communities. HOPE outperformed the above four baselines, but the detected communities remained fragmented and incomplete. GAE identified several incomplete predefined communities, with some nodes misclassified across communities.

All experiments were conducted on a workstation equipped with an Intel i9-13900KF CPU (32 threads) and an NVIDIA RTX 4090 GPU. The computational efficiency of all seven methods was evaluated on the simulated datasets. For each network in simulated datasets, the running time of each method was recorded, and the final results were reported as the average running time over the 10 networks for each dataset. Table 1 summarizes the average running time of each method in seconds. Overall, the statistical community detection methods, including Louvain, Leiden, and ScLeiden, exhibited the highest computational efficiency, consistently achieving the shortest running times. Among the embedding-based methods, HOPE was faster than node2vec, reflecting its lower computational complexity. GAE required substantially more computation time, owing to the computational cost of model training. USGT required the most computation time, as it employs a Transformer-based architecture involving global attention computation.

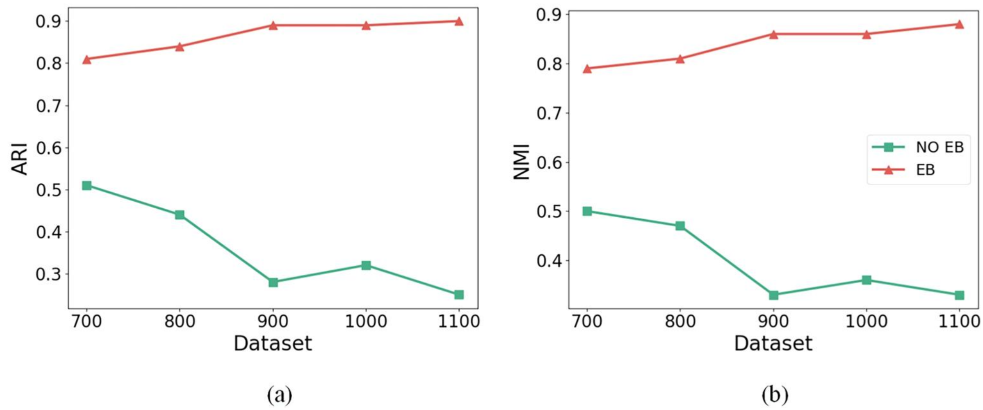


Figure 9. Comparison of community detection performance with and without embedding: (a) ARI; (b) NMI.

4.4. Ablation study

To systematically assess the contribution of the node embedding module and individual model components, a set of ablation experiments was conducted on the simulated datasets. Specifically, we investigated five aspects: (1) the effect of the node embedding module; (2) the effect of spatial encoding; (3) the effect of attention bias; (4) the impact of different data masking rates; and (5) the impact of different clustering methods. All models were trained using identical parameter settings to ensure a fair comparison.

4.4.1. Effect of the node embedding module

To evaluate the effectiveness of the proposed node embedding module, we compared the community detection performance of applying the SNN clustering method to node representations before and after embedding. For convenience, the former is referred to as *without embedding* (NO EB), and the latter as *with embedding* (EB). As shown in Figure 9, directly applying the clustering method to the raw node representations (i.e. the weighted adjacency matrix $\boldsymbol{M}^{w}$) resulted in poor performance, which further decreased with increasing dataset size. These results suggest that the proposed node embedding module effectively preserves the modular structure of the network, thereby facilitating more accurate community detection.

4.4.2. Effect of spatial encoding

As shown in Figure 10, we compared the model performance with and without spatial encoding. For convenience, we refer to the model with spatial encoding as SE and that without as NO SE. Without spatial encoding, the performance of the model decreased markedly, especially for smaller datasets. For instance, when $N = 700$, both ARI and NMI values fell below 0.45. These results suggest that the spatial encoding scheme effectively captures the relationships among spatial units, thereby enhancing the effectiveness of spatial community detection.

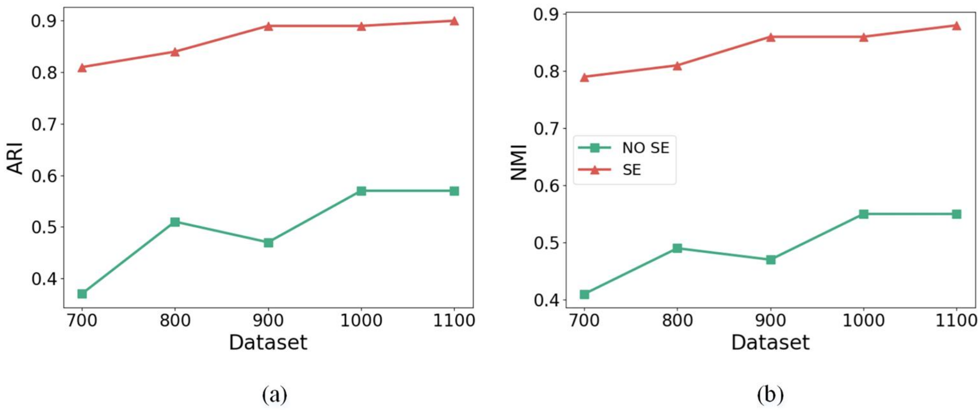


Figure 10. Performance of the proposed method with and without spatial encoding: (a) ARI; (b) NMI.

4.4.3. Effect of attention bias

We further investigated the impact of the modified attention mechanism, as illustrated in Figure 11. For convenience, we denote the model with attention bias as AB and that without as NO AB. The results indicated that the model with attention bias yielded consistently better performance than that without the bias. This suggests that incorporating both global and local information helps preserve the modular structure in node embeddings.

4.4.4. Impact of different data masking rates

The size of the subgraphs should be moderate: overly small subgraphs may lack sufficient information, whereas overly large ones can lead to overfitting. To determine an appropriate data masking rate, we varied it from 0.1 to 0.9, increasing by 0.1 at each step. The model performance across different data masking rates is presented in Figure 12. Although a masking rate of 0.3 yielded slightly better performance on some datasets, a masking rate of 0.5 was used for all simulation experiments and the case study. This choice retains sufficient nodes to capture local structural information while masking enough nodes for effective model training, thus ensuring stable performance across datasets of varying sizes.

When the masking rate was low (0.1–0.2), the performance was relatively poor. This may be because most nodes were retained, resulting in overfitting to local structures. Conversely, at high masking rates (0.7–0.9), performance also declined, likely due to excessive node removal, which reduced the information available in the sampled subgraphs for model training.

4.4.5. Impact of different clustering methods

To further investigate the impact of clustering methods, we compared the results obtained using the SNN clustering method with those obtained using two widely used clustering algorithms: K-means (MacQueen 1967) and fuzzy c-means (FCM, Bezdek *et al.* 1984). Both baseline methods require the number of clusters to be specified in advance. In the simulated datasets, the number of clusters was set to the

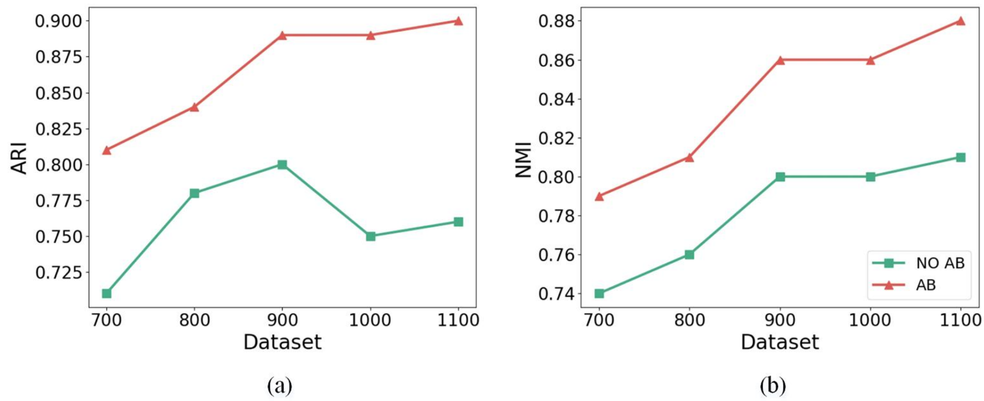


Figure 11. Performance of the proposed method with and without attention bias: (a) ARI; (b) NMI.

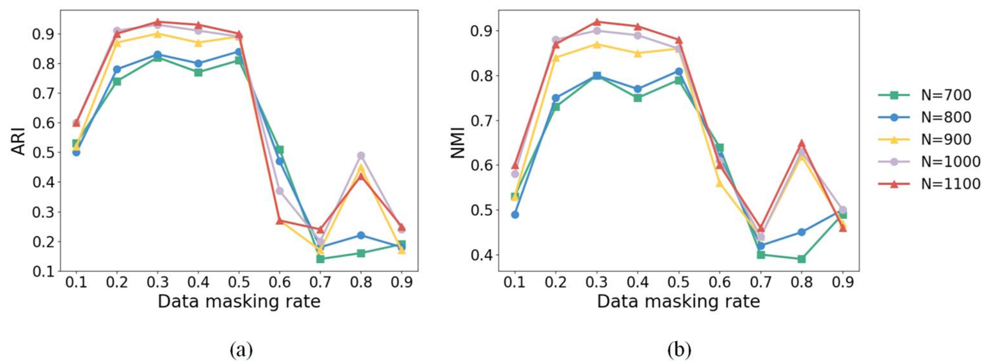


Figure 12. Performance of the proposed method across different masking rates: (a) ARI; (b) NMI.

number of predefined communities plus one, with all noise nodes assigned to an additional cluster. As shown in Figure 13, the SNN-based clustering method consistently outperformed the other methods across all datasets, demonstrating its superior ability to identify communities from the learned node embeddings.

5. Case study on Beijing taxi GPS trajectories

5.1. Study area and dataset

The study area was located within the Sixth Ring Road of Beijing, including a total of 1,451 traffic analysis zones (TAZs). GPS trajectory data from over 30,000 taxis were collected on May 8, 2016 (Sunday), and May 9, 2016 (Monday). Each trajectory record contained the taxi ID, occupancy status, longitude, latitude, and timestamp. After data cleaning, 233,170 OD pairs were obtained on Sunday and 322,602 on Monday. These OD pairs were mapped onto TAZs using the ray intersection method (Taylor 1994) to construct spatially embedded networks. To further analyze the identified communities, 21 locations were highlighted, as shown in Figure 14, with detailed descriptions provided in Table 2.

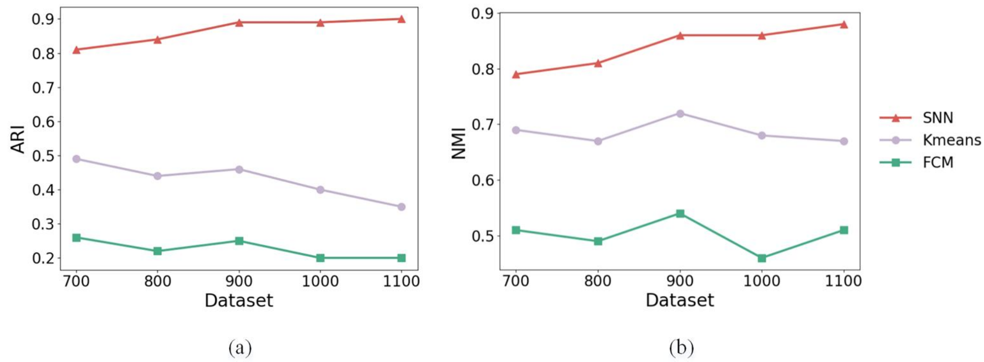


Figure 13. Performance of different clustering methods: (a) ARI; (b) NMI.

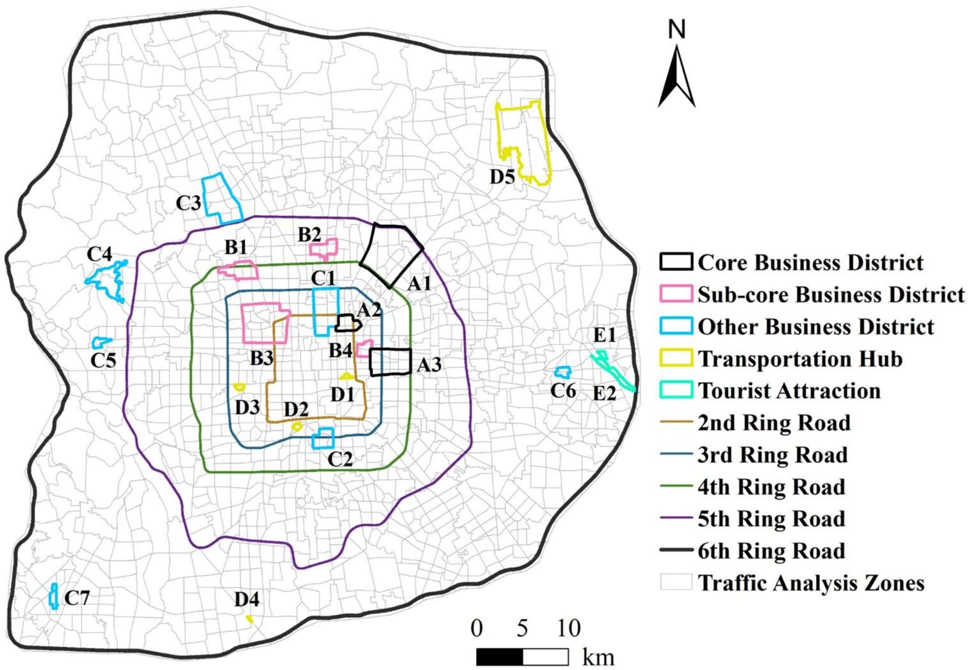


Figure 14. Overview of the study area.

Table 2. Description of the locations highlighted in Figure 14.

Type of locations	Name of locations
Core Business District	A1: Wangjing, A2: Dongzhimen, A3: Beijing CBD
Sub-core Business District	B1: Zhongguancun, B2: Yaao, B3: Xizhimen, B4: Chaowai
Other Business District	C1: Andingmen, C2: Muxiyuan, C3: Shangdi, C4: Xishan, C5: Pingguoyuan, C6: Tongzhou Wanda Plaza, C7: Fangshan
Transportation Hub	D1: Beijing Railway Station, D2: Beijingnan Railway Station, D3: Beijingxi Railway Station, D4: Huangcun Railway Station, D5: Beijing Capital International Airport
Tourist attraction	E1: Moon River Resort, E2: Tongzhou Canal Park

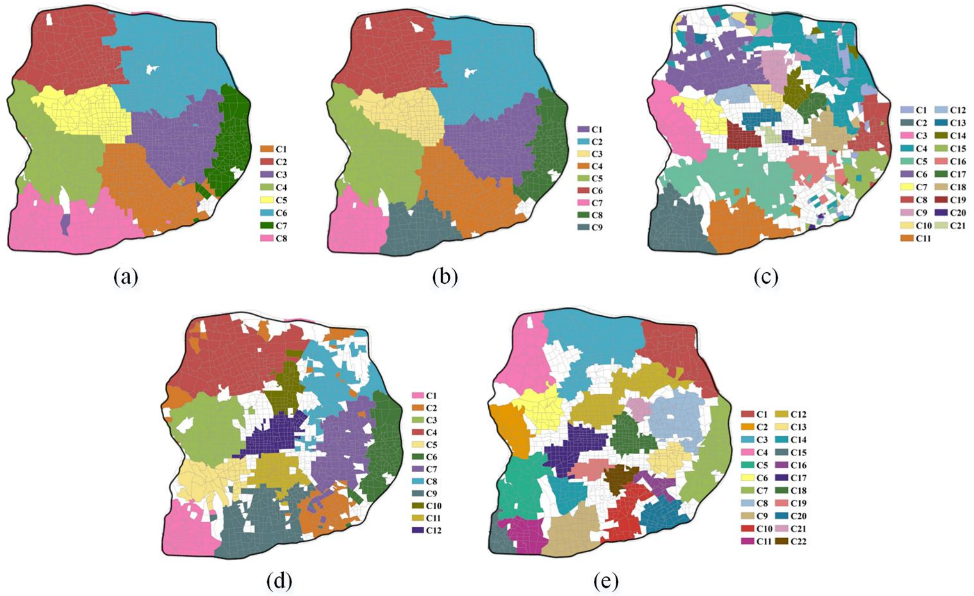


Figure 15. Spatial communities identified by different methods on Sunday: (a) Louvain; (b) ScLeiden; (c) HOPE; (d) GAE; (e) the proposed method.

For comparison, Louvain, ScLeiden, HOPE, and GAE, which were previously evaluated in the simulation experiments, were also applied in the case study as representatives of statistical and embedding-based approaches. The parameter settings were kept consistent with those used in the simulation experiments.

5.2. Identification of spatial communities

Figures 15 and 16 illustrate the identified spatial communities using Louvain, ScLeiden, HOPE, GAE, and the proposed method, on Sunday (weekend) and Monday (weekday), respectively. For clarity and readability, only communities with more than 15 members are presented. The traditional statistical methods, namely Louvain and ScLeiden, identified fewer but larger communities, and their spatial distributions were highly similar (as shown in Figures 15(a,b) and 16(a,b)). This similarity can be attributed to the fact that both methods optimize modularity as their objective function. Notably, both Louvain and ScLeiden identified geographically contiguous communities, even though Louvain does not explicitly incorporate spatial constraints during community detection. This may be due to the characteristics of the taxi trajectory data used in this study, in which most trips were short-distance, resulting in inherently spatially localized interactions. However, since every node in the mobility network was assigned to a community—even in areas with weakly connected interactions—these methods may identify spurious communities.

In contrast, HOPE divided the study area into a much larger number of smaller, spatially scattered communities (Figures 15(c) and 16(c)). These communities often lacked spatial continuity, with many being fragmented into multiple disconnected regions, thereby reducing their interpretability. The communities detected by GAE were

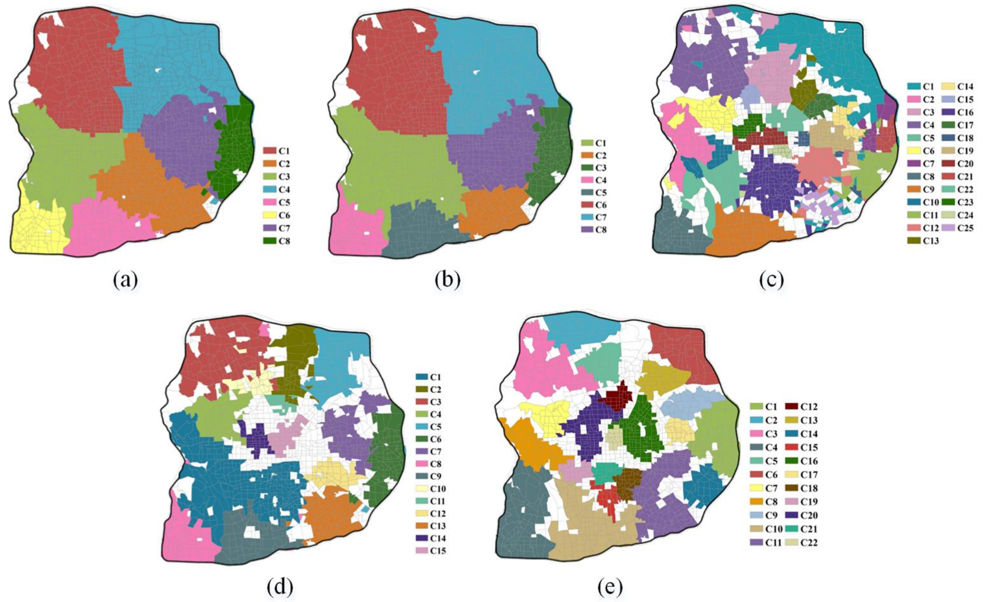


Figure 16. Spatial communities identified by different methods on Monday: (a) Louvain; (b) ScLeiden; (c) HOPE; (d) GAE; (e) the proposed method.

generally more geographically contiguous than those identified by HOPE; however, discontinuities remained in some cases, such as community C2 in Figure 15(d). By comparison, the proposed method captured finer-grained spatial structures while preserving spatial continuity in the resulting communities (Figures 15(e) and 16(e)). These results highlight its ability to capture more compact and interpretable spatial patterns.

To quantitatively evaluate the performance of these methods, the average subgraph density (ASD) of the identified communities was used (Wasserman and Faust 1994, Liu *et al.* 2019). ASD is defined as the average ratio of the actual number of edges to the maximum possible number of edges within the detected communities. In spatially embedded graphs constructed from sparse and noisy human mobility data, not all nodes necessarily belong to a community. However, most existing evaluation metrics (e.g. modularity) assume that every node is assigned to a community, making them unsuitable for evaluating such graphs. By contrast, ASD focuses on measuring the internal connectivity of the detected communities while being less sensitive to noise nodes, and is therefore better suited for evaluating communities in human mobility data.

Let $\mathbf{C} = \{C_1, C_2, \ldots, C_K\}$ denote the discovered communities. The ASD of $\mathbf{C}$ is computed as:

$$\mathrm{ASD}(\mathbf{C}) = \frac{1}{K}\sum\nolimits_{k=1}^{K} \frac{2W_k}{N_k(N_k-1)} \tag{11}$$

where K denotes the number of discovered communities, W_k represents the number of OD pairs within community C_k, and N_k represents the number of nodes in C_k. A higher ASD value indicates stronger internal connectivity within the communities.

Table 3. ASD values of the discovered communities using different methods.

	Louvain	ScLeiden	HOPE	GAE	USGT
Sunday	1.39	1.59	0.20	1.66	2.91
Monday	1.73	1.62	0.36	2.86	3.50

As shown in Table 3, the proposed method consistently achieved the highest ASD values on both Monday (3.50) and Sunday (2.91), indicating that the detected communities exhibited strong internal connectivity. In contrast, HOPE yielded the poorest performance, with substantially lower ASD values, suggesting weak internal OD interactions within the detected communities.

The ASD values obtained by Louvain and ScLeiden were comparable to each other. Although both methods outperformed HOPE, their ASD values remained notably lower than those of the proposed method. This is likely because both methods assign all nodes to communities, even in areas with weakly connected interactions, which reduces overall internal connectivity. GAE achieved suboptimal performance, indicating that it can identify communities with relatively dense internal interactions but remains less effective than the proposed method in detecting more cohesive communities. Across all methods, ASD values were higher on Monday (weekday) than on Sunday (weekend), consistent with expectations, given the increased travel demand from commuting behavior.

The spatial communities derived from taxi trajectories provide valuable insights into the polycentric structure of Beijing. To further analyze this structure, the identified spatial communities were overlaid with the locations highlighted in Figure 14, as shown in Figure 17. The identified spatial communities correspond closely to business districts (e.g. Wangjing, Dongzhimen, Beijing CBD, Chaowai, and Yaao) and key transportation hubs (e.g. Beijing Railway Station and Beijing Capital International Airport), which are characterized by high attractiveness and intensive spatial interactions. These communities reveal polycentric urban structures and offer a useful basis for validating and optimizing urban planning boundaries (Zhong *et al.* 2014, Wang *et al.* 2020). Moreover, these communities can reveal heterogeneous traffic interactions across urban space, providing valuable insights that support traffic control and traffic forecasting applications (Lu *et al.* 2018, Liu *et al.* 2019).

We further compared the spatial communities identified on Sunday and Monday (as shown in Figure 17), observing notable differences in their distributions. For instance, in Region I, two communities identified on Sunday merged into a single community on Monday. Region I contains several core business districts, including Wangjing, Dongzhimen, and the Beijing CBD. On Sunday, Wangjing and the Dongzhimen–Beijing CBD area were identified as two separate communities, as OD interactions between the mixed residential-employment Wangjing area and the employment-oriented Dongzhimen–CBD area were relatively weak. On Monday, commuting activities strengthened spatial interactions within Region I, leading to the formation of a larger community. In contrast, in Region II, a community identified on Sunday was split into two on Monday. Region II includes Tongzhou Wanda Plaza, Moon River Resort, and Tongzhou Canal Park. On Sunday, people tended to visit these locations for shopping and entertainment, which strengthened interactions within this area and led to the formation of a larger community. On Monday, however, the

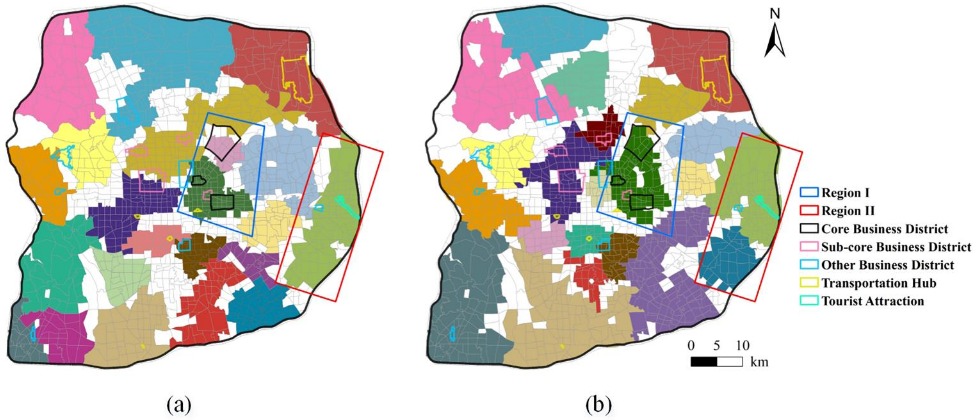


Figure 17. Overlay of the identified spatial communities and the highlighted locations: (a) Sunday; (b) Monday.

movements of residents were constrained by their work locations, weakening interactions within this region.

6. Discussion

Simulation experiments demonstrated that the proposed method outperformed six representative methods. In the case study, the spatial communities detected by the proposed method were more geographically contiguous and exhibited stronger internal connectivity compared with those identified by Louvain, ScLeiden, HOPE, and GAE. The main reasons are as follows.

1. Statistical methods (i.e. Louvain, Leiden, and ScLeiden) identify communities by optimizing modularity. These methods assign all nodes to communities, making it difficult to distinguish noise from true community structure. Moreover, they rely on the statistical characteristics of interactions, which limits their ability to detect spatial communities accurately in sparse and noisy human mobility data. Specifically, ScLeiden, a spatially constrained extension of Leiden, incorporates spatial constraints to detect spatially contiguous communities. Nevertheless, it still assigns all nodes to communities, which may lead to the identification of spurious communities in sparsely connected regions.
2. Embedding-based methods differ in their design objectives. node2vec preserves the local structure of nodes by generating sequences through random walks, whereas HOPE captures higher-order proximities. GAE learns node embeddings via a GAT encoder that aggregates neighbor information with attention weights. However, these embeddings neither guarantee the preservation of the modular structure nor account for spatial characteristics. Consequently, in the case study, the communities identified by HOPE were highly fragmented and weakly connected, while those identified by GAE, though more contiguous, still exhibited discontinuities in certain regions.

3. For the proposed method, the spatial encoding scheme effectively encodes spatial relationships among units. The structure-aware self-attention mechanism captures both global and local node information, which helps preserve community structures in the embedding space. Moreover, a data masking strategy combined with a composite loss function enables unsupervised training. Collectively, these components enhance the ability of the proposed model to identify spatial community structures in sparse and noisy mobility data. Specifically, the learned embeddings are dense and encode rich structural information, helping to mitigate the limitations imposed by network sparsity. In addition, noise nodes that are weakly connected to any community tend to be separated from community structures in the embedding space. Finally, the SNN clustering method can effectively distinguish community structures from noise, further reducing the influence of noise.

The case study based on taxi trajectories in Beijing further validated the effectiveness of the proposed model. Compared with four comparative methods, it more effectively identified spatial communities that were spatially contiguous and exhibited strong internal connectivity. As discussed in Section 5.2, the spatial communities detected using the proposed approach facilitate the identification of the polycentric urban structure, providing valuable insights for urban planning and traffic management.

7. Conclusion

This study developed a novel framework for detecting spatial communities in movement data based on deep graph embedding. By integrating both global and local information, the proposed method learns node embeddings that effectively preserve the modular structure of the network. To incorporate spatial constraints into the model, we designed a spatial encoding scheme based on spatial proximity between spatial units. Furthermore, the model was trained in an unsupervised manner by applying a data masking strategy and optimizing a composite loss function. Experiments on a series of synthetic datasets demonstrated the superiority of the proposed approach compared with six representative algorithms. The case study using taxi trajectories in Beijing further validated the effectiveness of the proposed approach. The identified communities not only enhance the understanding of mobility patterns in taxi trajectories, but also provide insights into the polycentric urban structure.

There are several limitations that need to be further investigated. First, training the proposed model is time- and space-consuming. The quadratic complexity of the self-attention mechanism limits its application to large graphs. Therefore, future work should focus on improving efficiency by leveraging high-performance computing and distributed storage techniques. Moreover, this study focused on detecting non-overlapping spatial communities. Although overlapping communities can, in principle, be identified using fuzzy clustering approaches, existing methods often struggle to identify them accurately and completely in high-dimensional, non-uniform embedding spaces. Extending the proposed framework to effectively identify overlapping spatial communities will be an important direction for future work.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

AI disclosure statement

During the preparation of this manuscript, the authors used ChatGPT (OpenAI, GPT-4) to assist with language polishing and grammar improvement. The tool was used solely to improve clarity and grammar. All ideas, methodologies, experiments, and conclusions were developed and verified by the authors. The use of the tool was intended to improve readability and language quality.

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Data and codes availability statement

The codes and instructions that support the findings of this study are available on 'figshare.com', with the identifier at the private link: https://doi.org/10.6084/m9.figshare.31968861. The taxi trajectory data cannot be made publicly available due to third party restrictions. Simulated data are shared at the link to demonstrate how the codes work.

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